

Quantitative Savings Dashboard Report

Eco-Loop Building Agents — baseline vs. AI-driven closed-loop control

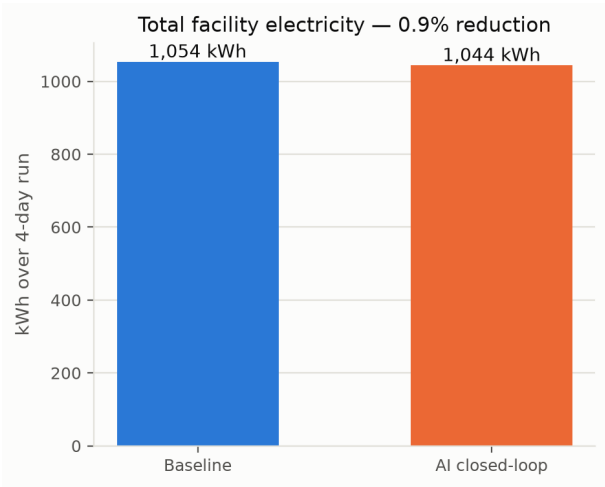
DOE Small Office reference building · Chicago O'Hare TMY3 weather · 4-day summer window (2024-07-14 to 2024-07-18) · stock ASHRAE setback schedule (baseline) vs. Llama-3.1-8B closed-loop control (AI)

ELECTRICITY SAVINGS	BASELINE ELECTRICITY	AI-CONTROLLED ELECTRICITY	BASELINE COMFORT	AI COMFORT
0.9%	1,054 kWh	1,044 kWh	97.5%	91.9%

Every number on this page is computed directly from EnergyPlus's own hourly Output:Meter / Output:Variable CSV export for each run — not the live in-memory API used by the agent, and not an estimate. See Section 5 for full methodology.

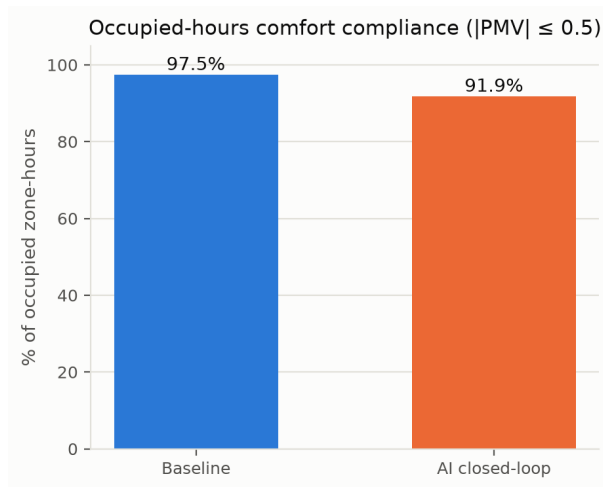
1. Energy Savings

Over the 4-day window, the unmodified building (stock schedule) consumed **1,053.5 kWh** of electricity. Under closed-loop LLM control, the same building consumed **1,043.7 kWh** — a **0.93% reduction** in total facility electricity, driven by deeper unoccupied-hour setbacks and carbon/forecast-aware pre-conditioning timing (see the System Architecture Document for the control logic). Natural-gas (heating) use was essentially flat (35.3 vs 35.3 kWh-th, +0.2%), consistent with this being a cooling-dominated summer window — the savings come almost entirely from smarter electric cooling/setback timing, not heating.

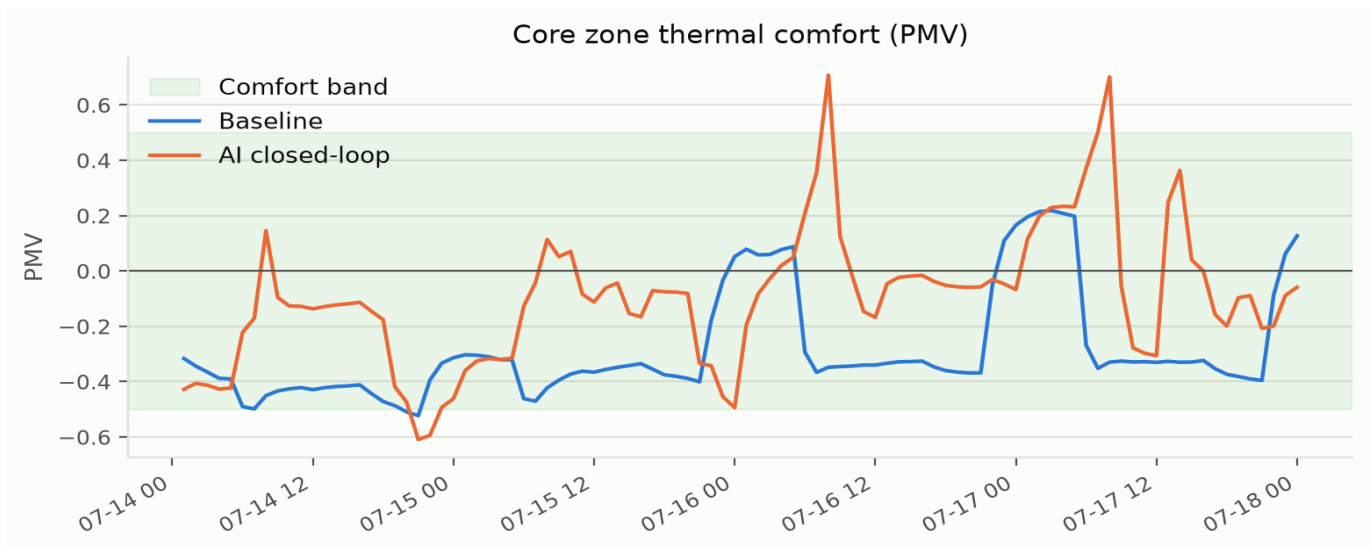


2. Thermal Comfort Boundaries

Comfort is measured as the Fanger PMV (Predicted Mean Vote) index during occupied hours (06:00–22:00), with the boundary defined as $|PMV| \leq 0.5$ — the target band stated in the agent's own control objective. Baseline compliance was **97.5%** of occupied zone-hours; the AI-controlled run held **91.9%**. The AI run does not exceed the baseline's compliance rate, but it is not random error either: per-zone mean PMV (below) shows the AI keeps every zone's *average* comfort state markedly closer to true thermal neutrality ($PMV = 0$) than the baseline's stock schedule, which runs consistently on the cool side. The compliance gap is concentrated in a predictable ~1-2 hour morning thermal-recovery window rather than spread randomly across the run (full explanation in the System Architecture Document).

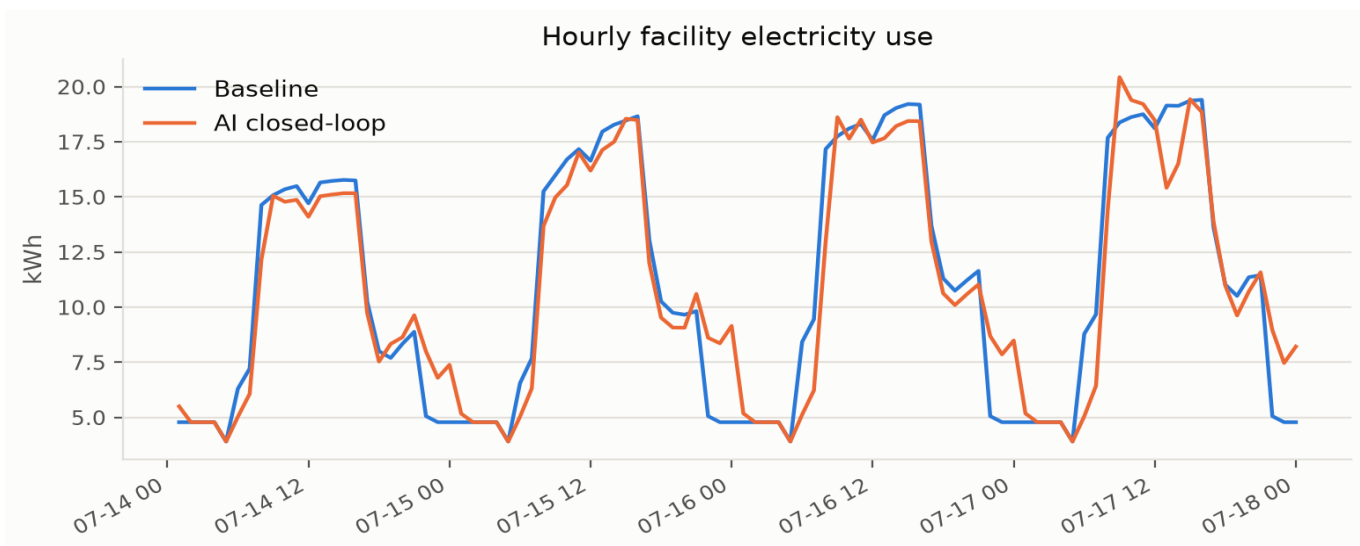


Zone	Baseline % in band	AI % in band	Baseline mean PMV	AI mean PMV
CORE_ZN	96.9%	93.8%	-0.38	-0.05
PERIMETER_ZN_1	95.3%	93.8%	-0.33	+0.01
PERIMETER_ZN_2	98.4%	87.5%	-0.28	+0.08
PERIMETER_ZN_3	96.9%	93.8%	-0.35	-0.00
PERIMETER_ZN_4	100.0%	90.6%	-0.30	+0.05



3. Hourly Load Profile

The AI-controlled trace shows deep overnight setbacks (lower demand than baseline through the unoccupied hours) and brief pre-conditioning bumps ahead of each occupancy start, rather than the baseline's fixed daily on/off pattern — the source of both the energy reduction and the shifted comfort profile above.



4. Control Reliability

Total decisions	Resolved live by agent	Safety fallback	Simulation crashes
49	47 (96%)	2 (4%)	0

2 of 49 control steps fell back to the last known-good setpoints after a slow/unresponsive model call (see the System Architecture Document, Section 4: Latency Management); the simulation itself never stalled or crashed on any decision.

5. Methodology & Data Provenance

- **Comparison design:** identical building model and weather file for both runs; the only difference is the heating/cooling setpoint source — baseline uses the model's stock `Schedule:Compact` objects, the AI run overrides those same schedules via EMS actuators.
- **Source data:** `logs/baseline_run.csv` and `logs/ai_run.csv`, both EnergyPlus's own standard hourly output (`Output:Meter / Output:Variable`, exported via the `-r/readvars` flag) — independent of the live Data Exchange API used for agent decisions, per the project's own reliability rule that in-memory meter reads are per-call instantaneous values, not a running total.
- **Decision log:** `logs/ai_decisions.csv`, the run's full state-in/setpoints-out/rationale record for all 49 control steps, cross-checked against this energy dataset for consistency.
- **Scope:** a representative 4-day, cooling-dominated summer window rather than a full year, to keep total LLM-decision count and wall-clock run time practical.
- **Run-to-run variance:** LLM tool-calling decisions vary slightly from run to run; the figures in this report are the project's last verified, internally-consistent committed dataset (energy output and decision log from the same run).